

Examples:**Ex. (1):** Find the gradient and Laplacian of the scalar field $\phi = xy^2z^3$?**Solution:**

$$\text{grad } \phi = \nabla \phi = i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z} = y^2 z^3 i + 2xyz^3 j + 3xy^2 z^2 k$$

$$\nabla^2 \phi = \nabla \cdot \nabla \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 2xz^3 + 6xy^2 z$$

Ex. (2): Find the divergence and curl of the vector field $\mathbf{A} = x^2 y^2 z^2 i + y^2 z^2 j + x^2 z^2 k$?**Solution:**

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 2(xy^2 z^2 + yz^2 + x^2 z)$$

$$\begin{aligned} \text{curl } \mathbf{A} = \nabla \times \mathbf{A} &= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 y^2 z^2 & y^2 z^2 & x^2 z^2 \end{vmatrix} \\ &= -2[y^2 z i + (xz^2 - x^2 y^2 z) j + x^2 y z^2 k] \end{aligned}$$

Ex. (3): Find (a) ∇r , (b) $\nabla \cdot \mathbf{r}$ (c) $\nabla \times \mathbf{r}$?**Solution:** $\mathbf{r} = xi + yj + zk$, $r = |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$ Then

$$\begin{aligned} \text{(a) } \nabla r &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \sqrt{x^2 + y^2 + z^2} = \\ &= \frac{1}{2} \left\{ i \frac{2x}{\sqrt{x^2 + y^2 + z^2}} + j \frac{2y}{\sqrt{x^2 + y^2 + z^2}} + k \frac{2z}{\sqrt{x^2 + y^2 + z^2}} \right\} \\ &= \frac{xi + yj + zk}{\sqrt{x^2 + y^2 + z^2}} = \frac{\mathbf{r}}{r} = \hat{r} \end{aligned}$$

$$(b) \nabla \cdot \mathbf{r} = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (xi + yj + zk)$$

$$= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3$$

$$(c) \nabla \times \mathbf{r} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \left(\frac{\partial z}{\partial y} - \frac{\partial y}{\partial z} \right) i + \left(\frac{\partial x}{\partial z} - \frac{\partial z}{\partial x} \right) j + \left(\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y} \right) k = 0$$

Ex. (4): Find gradient ϕ if (a) $\phi = \ln r$, (b) $\phi = 1/r$ (c) $\phi = r^n$?

Solution: (a) $\mathbf{r} = xi + yj + zk$. Then $r = |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$ and $\phi = \ln r = \ln \sqrt{x^2 + y^2 + z^2}$.

$$\begin{aligned} \text{grad } \phi &= \nabla \phi = \nabla \ln \sqrt{x^2 + y^2 + z^2} \\ &= \frac{1}{2} \left\{ i \frac{\partial}{\partial x} \ln(x^2 + y^2 + z^2) + j \frac{\partial}{\partial y} \ln(x^2 + y^2 + z^2) + k \frac{\partial}{\partial z} \ln(x^2 + y^2 + z^2) \right\} \\ &= \frac{1}{2} \left\{ i \frac{2x}{x^2 + y^2 + z^2} + j \frac{2y}{x^2 + y^2 + z^2} + k \frac{2z}{x^2 + y^2 + z^2} \right\} = \frac{xi + yj + zk}{x^2 + y^2 + z^2} \\ &= \frac{\mathbf{r}}{r^2} \end{aligned}$$

$$\begin{aligned} (b) \text{ grad } \phi &= \nabla \phi = \nabla 1/r = \nabla \frac{1}{\sqrt{x^2 + y^2 + z^2}} = \nabla \left\{ (x^2 + y^2 + z^2)^{-1/2} \right\} \\ &= i \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-1/2} + j \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{-1/2} + k \frac{\partial}{\partial z} (x^2 + y^2 + z^2)^{-1/2} \\ &= i \left\{ -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2x) \right\} + j \left\{ -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2y) \right\} \\ &\quad + k \left\{ -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2z) \right\} \\ &= \frac{-xi - yj - zk}{(x^2 + y^2 + z^2)^{3/2}} = -\frac{\mathbf{r}}{r^3} \equiv -\frac{\hat{\mathbf{r}}}{r^2} \end{aligned}$$

$$\begin{aligned}
\text{(c) } \text{grad } \phi &= \nabla \phi = \nabla r^n = \nabla (\sqrt{x^2 + y^2 + z^2})^n = \nabla (x^2 + y^2 + z^2)^{n/2} \\
&= i \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{n/2} + j \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{n/2} + k \frac{\partial}{\partial z} (x^2 + y^2 + z^2)^{n/2} \\
&= i \left\{ \frac{n}{2} (x^2 + y^2 + z^2)^{n/2-1} (2x) \right\} + j \left\{ \frac{n}{2} (x^2 + y^2 + z^2)^{n/2-1} (2y) \right\} \\
&\quad + k \left\{ \frac{n}{2} (x^2 + y^2 + z^2)^{n/2-1} (2z) \right\} \\
&= n (x^2 + y^2 + z^2)^{n/2-1} (xi + yj + zk) = n (r^2)^{n/2-1} \mathbf{r} = n r^{n-2} \mathbf{r} = n r^{n-1} \hat{\mathbf{r}}
\end{aligned}$$

Ex. (5): Find the directional derivative of $\Phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2i - j - 2k$?

Solution: $\nabla \phi = \nabla (x^2yz + 4xz^2) = i(2xyz + 4z^2) + jx^2z + k(x^2y + 8xz)$

$$= 8i - j - 10k \quad \text{at } (1, -2, -1)$$

The unit vector in the direction of $2i - j - 2k$ is

$$\hat{a} = \frac{\mathbf{a}}{a} = \frac{2i - j - 2k}{\sqrt{(2)^2 + (-1)^2 + (-2)^2}} = \frac{1}{3}(2i - j - 2k)$$

Then the required directional derivative is

$$\nabla \phi \cdot \hat{a} = (8i - j - 10k) \cdot \left(\frac{1}{3}(2i - j - 2k) \right) = \frac{1}{3}(16 + 1 + 20) = \frac{37}{3}$$

Since this is positive, Φ is increasing in this direction.

Ex. (6): Given $\Phi = 2x^3 y^2 z^4$, Find $\text{div grad } \Phi$?

Solution: $\nabla \phi = \nabla (2x^3 y^2 z^4) = 6x^2 y^2 z^4 i + 4x^3 y z^4 j + 8x^3 y^2 z^3 k$

$$\therefore \text{div grad } \phi = \nabla \cdot \nabla \phi$$

$$\begin{aligned}
&= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (6x^2 y^2 z^4 i + 4x^3 y z^4 j + 8x^3 y^2 z^3 k) \\
&= 12x y^2 z^4 + 4x^3 z^4 + 24x^3 y^2 z^2
\end{aligned}$$

Ex. (7): Prove that $\nabla^2 \frac{1}{r} = 0$?

Solution: $\mathbf{r} = xi + yj + zk$, $r = |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$

$$\nabla^2 \frac{1}{r} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right)$$

$$\frac{\partial}{\partial x} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) = -x(x^2 + y^2 + z^2)^{-3/2}$$

$$\begin{aligned} \frac{\partial^2}{\partial x^2} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) &= \frac{\partial}{\partial x} \{ -x(x^2 + y^2 + z^2)^{-3/2} \} \\ &= 3x^2(x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2} \\ &= \frac{2x^2 - y^2 - z^2}{(x^2 + y^2 + z^2)^{5/2}} \end{aligned}$$

Similarly,

$$\frac{\partial^2}{\partial y^2} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) = \frac{2y^2 - z^2 - x^2}{(x^2 + y^2 + z^2)^{5/2}}$$

$$\frac{\partial^2}{\partial z^2} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) = \frac{2z^2 - x^2 - y^2}{(x^2 + y^2 + z^2)^{5/2}}$$

$$\begin{aligned} \therefore \nabla^2 \frac{1}{r} &= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) \\ &= \frac{2x^2 - y^2 - z^2}{(x^2 + y^2 + z^2)^{5/2}} + \frac{2y^2 - z^2 - x^2}{(x^2 + y^2 + z^2)^{5/2}} + \frac{2z^2 - x^2 - y^2}{(x^2 + y^2 + z^2)^{5/2}} \\ &= 0 \end{aligned}$$

Ex. (8): Prove that $\nabla \cdot \frac{\mathbf{r}}{r^3} = 0$?

Solution: $\mathbf{r} = xi + yj + zk$, $r = |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$

$$\begin{aligned}
\nabla \cdot \frac{\mathbf{r}}{r^3} &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot \left(\frac{xi + yj + zk}{(x^2 + y^2 + z^2)^{3/2}} \right) \\
&= \frac{\partial}{\partial x} \left(\frac{x}{(x^2 + y^2 + z^2)^{3/2}} \right) + \frac{\partial}{\partial y} \left(\frac{y}{(x^2 + y^2 + z^2)^{3/2}} \right) + \frac{\partial}{\partial z} \left(\frac{z}{(x^2 + y^2 + z^2)^{3/2}} \right) \\
\frac{\partial}{\partial x} \left(\frac{x}{(x^2 + y^2 + z^2)^{3/2}} \right) &= \frac{(x^2 + y^2 + z^2)^{3/2} - x(3x(x^2 + y^2 + z^2)^{1/2})}{(x^2 + y^2 + z^2)^3} \\
&= \frac{-2x^2 + y^2 + z^2}{(x^2 + y^2 + z^2)^{5/2}}
\end{aligned}$$

Similarly,

$$\begin{aligned}
\frac{\partial}{\partial y} \left(\frac{y}{(x^2 + y^2 + z^2)^{3/2}} \right) &= \frac{-2y^2 + z^2 + x^2}{(x^2 + y^2 + z^2)^{5/2}} \\
\frac{\partial}{\partial z} \left(\frac{z}{(x^2 + y^2 + z^2)^{3/2}} \right) &= \frac{-2z^2 + x^2 + y^2}{(x^2 + y^2 + z^2)^{5/2}}
\end{aligned}$$

$$\therefore \nabla \cdot \frac{\mathbf{r}}{r^3} = 0 \quad , \quad \text{Hint: } \nabla \cdot \frac{\mathbf{r}}{r^3} = -\nabla^2 \frac{1}{r} = 0 \quad , \quad \text{for } r \neq 0$$

Ex. (9): Prove: (a) $\nabla \cdot (\mathbf{A} + \mathbf{B}) = \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B}$

(b) $\nabla \cdot (\phi \mathbf{A}) = (\nabla \phi) \cdot \mathbf{A} + \phi(\nabla \cdot \mathbf{A})$?

Solution: (a) Let $\mathbf{A} = A_1i + A_2j + A_3k$, $\mathbf{B} = B_1i + B_2j + B_3k$

$$\begin{aligned}
\nabla \cdot (\mathbf{A} + \mathbf{B}) &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot [(A_1 + B_1)i + (A_2 + B_2)j + (A_3 + B_3)k] \\
&= \frac{\partial}{\partial x} (A_1 + B_1) + \frac{\partial}{\partial y} (A_2 + B_2) + \frac{\partial}{\partial z} (A_3 + B_3) \\
&= \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} + \frac{\partial B_1}{\partial x} + \frac{\partial B_2}{\partial y} + \frac{\partial B_3}{\partial z}
\end{aligned}$$

$$\begin{aligned}
&= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (A_1 i + A_2 j + A_3 k) \\
&\quad + \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (B_1 i + B_2 j + B_3 k) \\
&= \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B}
\end{aligned}$$

$$\begin{aligned}
\text{(b) } \nabla \cdot (\phi \mathbf{A}) &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (\phi A_1 i + \phi A_2 j + \phi A_3 k) \\
&= \frac{\partial}{\partial x} (\phi A_1) + \frac{\partial}{\partial y} (\phi A_2) + \frac{\partial}{\partial z} (\phi A_3) \\
&= \frac{\partial \phi}{\partial x} A_1 + \phi \frac{\partial A_1}{\partial x} + \frac{\partial \phi}{\partial y} A_2 + \phi \frac{\partial A_2}{\partial y} + \frac{\partial \phi}{\partial z} A_3 + \phi \frac{\partial A_3}{\partial z} \\
&= \frac{\partial \phi}{\partial x} A_1 + \frac{\partial \phi}{\partial y} A_2 + \frac{\partial \phi}{\partial z} A_3 + \phi \left(\frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) \\
&= \left(i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z} \right) \cdot (A_1 i + A_2 j + A_3 k) \\
&\quad + \phi \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (A_1 i + A_2 j + A_3 k) \\
&= (\nabla \phi) \cdot \mathbf{A} + \phi (\nabla \cdot \mathbf{A})
\end{aligned}$$

Ex. (10): If $\mathbf{A} = x^2 y \mathbf{i} - 2xz \mathbf{j} + 2yz \mathbf{k}$, find $\text{curl curl } \mathbf{A}$?

Solution: $\text{curl curl } \mathbf{A} = \nabla \times (\nabla \times \mathbf{A})$

$$\begin{aligned}
&= \nabla \times \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 y & -2xz & 2yz \end{vmatrix} = \nabla \times [(2x + 2z) \mathbf{i} - (x^2 + 2z) \mathbf{k}] \\
&= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x + 2z & 0 & -(x^2 + 2z) \end{vmatrix} = (2x + 2) \mathbf{j}
\end{aligned}$$

Ex. (11): Prove: (a) $\nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}$

(b) $\nabla \times (\phi \mathbf{A}) = (\nabla \phi) \times \mathbf{A} + \phi(\nabla \times \mathbf{A})$?

Solution: (a) Let $\mathbf{A} = A_1i + A_2j + A_3k$, $\mathbf{B} = B_1i + B_2j + B_3k$

$$\begin{aligned}
 \nabla \times (\mathbf{A} + \mathbf{B}) &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times [(A_1 + B_1)i + (A_2 + B_2)j + (A_3 + B_3)k] \\
 &= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 + B_1 & A_2 + B_2 & A_3 + B_3 \end{vmatrix} \\
 &= \left[\frac{\partial}{\partial y} (A_3 + B_3) - \frac{\partial}{\partial z} (A_2 + B_2) \right] i + \left[\frac{\partial}{\partial z} (A_1 + B_1) - \frac{\partial}{\partial x} (A_3 + B_3) \right] j \\
 &\quad + \left[\frac{\partial}{\partial x} (A_2 + B_2) - \frac{\partial}{\partial y} (A_1 + B_1) \right] k \\
 &= \left[\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right] i + \left[\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right] j + \left[\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right] k \\
 &\quad + \left[\frac{\partial B_3}{\partial y} - \frac{\partial B_2}{\partial z} \right] i + \left[\frac{\partial B_1}{\partial z} - \frac{\partial B_3}{\partial x} \right] j + \left[\frac{\partial B_2}{\partial x} - \frac{\partial B_1}{\partial y} \right] k \\
 &= \nabla \times \mathbf{A} + \nabla \times \mathbf{B}
 \end{aligned}$$

$$\text{(b) } \nabla \times (\phi \mathbf{A}) = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times (\phi A_1 i + \phi A_2 j + \phi A_3 k)$$

$$\begin{aligned}
 &= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \phi A_1 & \phi A_2 & \phi A_3 \end{vmatrix} \\
 &= \left[\frac{\partial}{\partial y} (\phi A_3) - \frac{\partial}{\partial z} (\phi A_2) \right] i + \left[\frac{\partial}{\partial z} (\phi A_1) - \frac{\partial}{\partial x} (\phi A_3) \right] j \\
 &\quad + \left[\frac{\partial}{\partial x} (\phi A_2) - \frac{\partial}{\partial y} (\phi A_1) \right] k
 \end{aligned}$$

$$\begin{aligned}
&= \left[\phi \frac{\partial A_3}{\partial y} + \frac{\partial \phi}{\partial y} A_3 - \phi \frac{\partial A_2}{\partial z} - \frac{\partial \phi}{\partial z} A_2 \right] i \\
&\quad + \left[\phi \frac{\partial A_1}{\partial z} + \frac{\partial \phi}{\partial z} A_1 - \phi \frac{\partial A_3}{\partial x} - \frac{\partial \phi}{\partial x} A_3 \right] j \\
&\quad + \left[\phi \frac{\partial A_2}{\partial x} + \frac{\partial \phi}{\partial x} A_2 - \phi \frac{\partial A_1}{\partial y} - \frac{\partial \phi}{\partial y} A_1 \right] k \\
&= \left[\phi \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) i + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) j + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) k \right] \\
&\quad + \left[\left(\frac{\partial \phi}{\partial y} A_3 - \frac{\partial \phi}{\partial z} A_2 \right) i + \left(\frac{\partial \phi}{\partial z} A_1 - \frac{\partial \phi}{\partial x} A_3 \right) j + \left(\frac{\partial \phi}{\partial x} A_2 - \frac{\partial \phi}{\partial y} A_1 \right) k \right] \\
&= \phi (\nabla \times \mathbf{A}) + \begin{vmatrix} i & j & k \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} = \phi (\nabla \times \mathbf{A}) + (\nabla \phi) \times \mathbf{A}
\end{aligned}$$

Ex. (12): Prove: **(a)** $\text{curl grad } \Phi = 0$, **(b)** $\text{div curl } \mathbf{A} = 0$?

Solution: **(a)** $\text{curl grad } \Phi = \nabla \times (\nabla \Phi)$

$$\begin{aligned}
&= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times \left(i \frac{\partial \Phi}{\partial x} + j \frac{\partial \Phi}{\partial y} + k \frac{\partial \Phi}{\partial z} \right) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \Phi}{\partial x} & \frac{\partial \Phi}{\partial y} & \frac{\partial \Phi}{\partial z} \end{vmatrix} \\
&= \left[\frac{\partial}{\partial y} \left(\frac{\partial \Phi}{\partial z} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \Phi}{\partial y} \right) \right] i + \left[\frac{\partial}{\partial z} \left(\frac{\partial \Phi}{\partial x} \right) - \frac{\partial}{\partial x} \left(\frac{\partial \Phi}{\partial z} \right) \right] j + \left[\frac{\partial}{\partial x} \left(\frac{\partial \Phi}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \Phi}{\partial x} \right) \right] k \\
&= \left[\frac{\partial^2 \Phi}{\partial y \partial z} - \frac{\partial^2 \Phi}{\partial z \partial y} \right] i + \left[\frac{\partial^2 \Phi}{\partial z \partial x} - \frac{\partial^2 \Phi}{\partial x \partial z} \right] j + \left[\frac{\partial^2 \Phi}{\partial x \partial y} - \frac{\partial^2 \Phi}{\partial y \partial x} \right] k = 0
\end{aligned}$$

(b) $\text{div curl } \mathbf{A} = \nabla \cdot (\nabla \times \mathbf{A})$

$$= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times (A_1 i + A_2 j + A_3 k)$$

$$\begin{aligned}
&= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} \\
&= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot \left[\left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) i + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) j + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) k \right] \\
&= \frac{\partial}{\partial x} \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \\
&= \frac{\partial^2 A_3}{\partial x \partial y} - \frac{\partial^2 A_2}{\partial x \partial z} + \frac{\partial^2 A_1}{\partial y \partial z} - \frac{\partial^2 A_3}{\partial y \partial x} + \frac{\partial^2 A_2}{\partial z \partial x} - \frac{\partial^2 A_1}{\partial z \partial y} = 0
\end{aligned}$$

Ex. (13): Prove: $\text{curl curl } \mathbf{A} = -\nabla^2 \mathbf{A} + \nabla(\nabla \cdot \mathbf{A})$?

Solution: $\text{curl curl } \mathbf{A} = \nabla \times (\nabla \times \mathbf{A}) = -\nabla^2 \mathbf{A} + \nabla(\nabla \cdot \mathbf{A})$

$$\begin{aligned}
&= \nabla \times (\nabla \times \mathbf{A}) = \nabla \times \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} \\
&= \nabla \times \left[\left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) i + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) j + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) k \right] \\
&= \nabla \times \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} & \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} & \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \end{vmatrix} \\
&= \left[\frac{\partial}{\partial y} \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) - \frac{\partial}{\partial z} \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \right] i \\
&\quad + \left[\frac{\partial}{\partial z} \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) - \frac{\partial}{\partial x} \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \right] j \\
&\quad + \left[\frac{\partial}{\partial x} \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \right] k \\
&= \left(-\frac{\partial^2 A_1}{\partial y^2} - \frac{\partial^2 A_1}{\partial z^2} \right) i + \left(-\frac{\partial^2 A_2}{\partial z^2} - \frac{\partial^2 A_2}{\partial x^2} \right) j + \left(-\frac{\partial^2 A_3}{\partial x^2} - \frac{\partial^2 A_3}{\partial y^2} \right) k \\
&\quad + \left(\frac{\partial^2 A_2}{\partial y \partial x} + \frac{\partial^2 A_3}{\partial z \partial x} \right) i + \left(\frac{\partial^2 A_3}{\partial z \partial y} + \frac{\partial^2 A_1}{\partial x \partial y} \right) j + \left(\frac{\partial^2 A_1}{\partial x \partial z} + \frac{\partial^2 A_2}{\partial y \partial z} \right) k
\end{aligned}$$

$$\begin{aligned}
&= \left(-\frac{\partial^2 A_1}{\partial x^2} - \frac{\partial^2 A_1}{\partial y^2} - \frac{\partial^2 A_1}{\partial z^2} \right) i + \left(-\frac{\partial^2 A_2}{\partial x^2} - \frac{\partial^2 A_2}{\partial y^2} - \frac{\partial^2 A_2}{\partial z^2} \right) j \\
&\quad + \left(-\frac{\partial^2 A_3}{\partial x^2} - \frac{\partial^2 A_3}{\partial y^2} - \frac{\partial^2 A_3}{\partial z^2} \right) k + \left(\frac{\partial^2 A_1}{\partial x^2} + \frac{\partial^2 A_2}{\partial y \partial x} + \frac{\partial^2 A_3}{\partial z \partial x} \right) i \\
&\quad + \left(\frac{\partial^2 A_1}{\partial x \partial y} + \frac{\partial^2 A_2}{\partial y^2} + \frac{\partial^2 A_3}{\partial z \partial y} \right) j + \left(\frac{\partial^2 A_1}{\partial x \partial z} + \frac{\partial^2 A_2}{\partial y \partial z} + \frac{\partial^2 A_3}{\partial z^2} \right) k \\
&= - \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (A_1 i + A_2 j + A_3 k) + i \frac{\partial}{\partial x} \left(\frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) \\
&\quad + j \frac{\partial}{\partial y} \left(\frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) + k \frac{\partial}{\partial z} \left(\frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) \\
&= -\nabla^2 \mathbf{A} + \nabla \left(\frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) = -\nabla^2 \mathbf{A} + \nabla(\nabla \cdot \mathbf{A})
\end{aligned}$$

Ex. (14): if $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, prove $\boldsymbol{\omega} = \frac{1}{2} \text{curl } \mathbf{v}$ where $\boldsymbol{\omega}$ is a constant vector?

Solution: $\mathbf{r} = xi + yj + zk$, $\boldsymbol{\omega} = \omega_1 i + \omega_2 j + \omega_3 k$

$$\begin{aligned}
\text{curl } \mathbf{v} &= \nabla \times \mathbf{v} = \nabla \times (\boldsymbol{\omega} \times \mathbf{r}) = \nabla \times \begin{vmatrix} i & j & k \\ \omega_1 & \omega_2 & \omega_3 \\ x & y & z \end{vmatrix} \\
&= \nabla \times [(\omega_2 z - \omega_3 y)i + (\omega_3 x - \omega_1 z)j + (\omega_1 y - \omega_2 x)k] \\
&= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \omega_2 z - \omega_3 y & \omega_3 x - \omega_1 z & \omega_1 y - \omega_2 x \end{vmatrix} = 2(\omega_1 i + \omega_2 j + \omega_3 k) = 2\boldsymbol{\omega} \\
\therefore \boldsymbol{\omega} &= \frac{1}{2} \text{curl } \mathbf{v}
\end{aligned}$$

Ex. (15): Which of the following are solenoidal or irrotational ?

$$(a) \mathbf{A} = 2xyz i + x^2 z j + x^2 y k \quad (b) \mathbf{B} = j + xy k$$

Solution:

$$(a) \nabla \cdot \mathbf{A} = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (2xyz i + x^2 z j + x^2 y k) = 2yz$$

$\therefore \nabla \cdot \mathbf{A} \neq 0 \rightarrow \mathbf{A}$ is not solenoidal.

$$\begin{aligned}\nabla \times \mathbf{A} &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times (2xyz \mathbf{i} + x^2z \mathbf{j} + x^2y \mathbf{k}) \\ &= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz & x^2z & x^2y \end{vmatrix} = 0\end{aligned}$$

$\therefore \nabla \times \mathbf{A} = \mathbf{0} \rightarrow \mathbf{A}$ is irrotational.

$$(b) \quad \nabla \cdot \mathbf{B} = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (j + xy \mathbf{k}) = 0$$

$\therefore \nabla \cdot \mathbf{B} = 0 \rightarrow \mathbf{B}$ is solenoidal.

$$\nabla \times \mathbf{B} = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times (j + xy \mathbf{k}) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 1 & xy \end{vmatrix} = (xi - yj)$$

$\therefore \nabla \times \mathbf{B} \neq \mathbf{0} \rightarrow \mathbf{B}$ is not irrotational.

Ex. (16): Prove that $\mathbf{F} = (y^2 - z^2 + 3yz - 2x)\mathbf{i} + (3xz + 2xy)\mathbf{j} + (3xy - 2xz + 2z)\mathbf{k}$ is both solenoidal and irrotational ?

$$\begin{aligned}\text{Solution: } \nabla \cdot \mathbf{F} &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (y^2 - z^2 + 3yz - 2x)\mathbf{i} + \\ &\quad (3xz + 2xy)\mathbf{j} + (3xy - 2xz + 2z)\mathbf{k} \\ &= -2 + 2x - 2x + 2 = 0\end{aligned}$$

$\therefore \nabla \cdot \mathbf{F} = 0 \rightarrow \mathbf{F}$ is solenoidal.

$$\begin{aligned}\nabla \times \mathbf{F} &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times (y^2 - z^2 + 3yz - 2x)\mathbf{i} + (3xz + 2xy)\mathbf{j} \\ &\quad + (3xy - 2xz + 2z)\mathbf{k} \\ &= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 - z^2 + 3yz - 2x & 3xz + 2xy & 3xy - 2xz + 2z \end{vmatrix} = i(3x - 3x) \\ &\quad + j((-2z + 3y) - (3y - 2z)) + k((3z + 2y) - (2y + 3z)) = 0\end{aligned}$$

$\therefore \nabla \times \mathbf{F} = \mathbf{0} \rightarrow \mathbf{F}$ is irrotational.